

Dynamical Calibration of a Metallicity-Dependent Stellar Mass–Luminosity Relation from Gaia Wide Binaries

Frozen model configuration: 28 September 2026

1 Scientific objective and modular architecture

We use modular inference. Stage 1a uses within-binary metallicity differences to estimate the relative JCAPS calibration. Conditional on that estimate, Stage 1b combines the component JCAPS measurements with anchor stars’ colour–magnitude diagram (CMD) and a hierarchical population distribution to infer a probability distribution for the shared metallicity of every binary. Stage 2 treats these distributions as fixed inputs, predicts the two component masses at every metallicity, and evaluates the wide-binary dynamical likelihood. The architecture is summarized in [figure 1](#).

2 Observables, latent quantities, and conventions

Binary systems are indexed by j , their two components by $k \in \{1, 2\}$, and metallicity nodes by q . A hat denotes an observed quantity. The symbol $\log_{10}$ is reserved for base-10 stellar-mass logarithms; unqualified log denotes the natural logarithm. “Dex” denotes an interval in a base-10 logarithm. The main quantities are listed in [table 1](#).

Table 1: Principal observables and latent or derived quantities.

Symbol	Meaning	Unit or range
$\mathcal{D}_{1a}$	Within-binary JCAPS differences used to estimate relative calibration parameters	data set
$\mathcal{D}_{1b}$	JCAPS and Gaia CMD data used for absolute calibration and system posteriors	data set
$\mathcal{D}_2$	Stage-2 component-magnitude and wide-binary dynamical data, with $\bar{P}_{jq}$ supplied from Stage 1b	data set
x_{jk}	Gaia absolute magnitude of component k	mag
C_{jk}	Dereddened colour $(G_{BP} - G_{RP})_0$	mag
$\hat{z}_{jk}$	Raw JCAPS XP metallicity measurement	dex
$\sigma_{z,jk}$	Reported, uncorrected JCAPS uncertainty	dex
Z_j	Shared true system metallicity on the PARSEC scale	dex
$\bar{P}_{jq}$	Stage-1b posterior probability mass at Z_q	$[0, 1]$
$g(x, Z)$	Final $\log_{10}(M/M_\odot)$ surface	dex
u_j	Observed $\Delta v_{\tan} \sqrt{r_\perp}$ amplitude	$\text{km s}^{-1} \sqrt{\text{AU}}$
$\sigma_{u,j}$	Component-wise Gaussian error scale entering the Rice model	$\text{km s}^{-1} \sqrt{\text{AU}}$
s_{jq}	Dynamical mass scale $\sqrt{M_{\text{tot},jq}/M_\odot}$	dimensionless

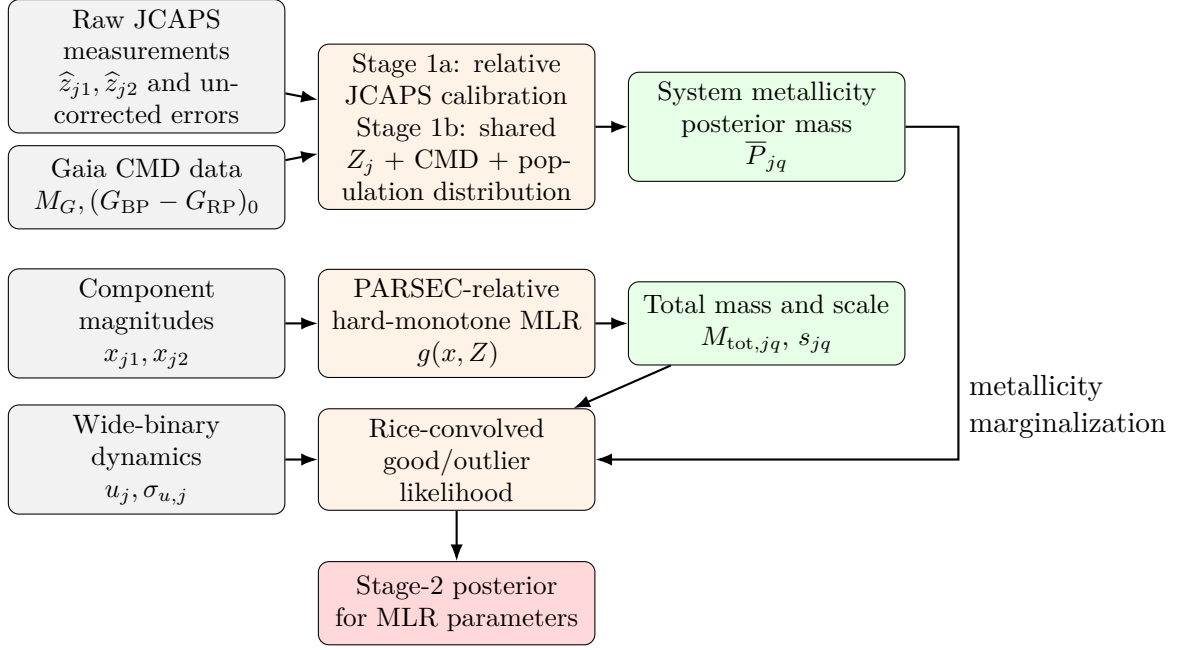


Figure 1: Scientific workflow. Stage 1a estimates the relative JCAPS calibration from within-binary differences. Conditional on those fixed estimates, Stage 1b returns a posterior probability mass over each shared system metallicity. Stage 2 propagates this uncertainty into the dynamical MLR fit but does not update either Stage 1 module. This one-way flow implements the modular cut.

The metallicity inputs are strictly `feh_jcaps_1`, `feh_jcaps_2`, `jc_sigma_m_h_1`, and `jc_sigma_m_h_2`. JCAPS infers stellar labels by inverting a forward model of Gaia XP spectra; the underlying framework and its training strategy are described by Li et al. (2026). The names above refer to columns in the analysis table and are not asserted to be published JCAPS catalogue field names. No binary-corrected, fitted, calibrated, or inflated metallicity column is used. Such columns already incorporate the equal-composition assumption that appears in our likelihood; reusing them would count that information twice and make the calibration circular.

3 Stage 1: relative and absolute metallicity calibration

3.1 Stage 1a: Relative JCAPS calibration

Before introducing the PARSEC CMD, the relative behavior of JCAPS is calibrated from the difference between the two stars in each binary. The adopted observation model assigns one independent Student- t state to each star,

$$\hat{z}_{jk} \mid Z_j \sim \mathcal{T}_\nu \left[Z_j + \delta_z + b_G(x_{jk}), \sqrt{\sigma_{z,jk}^2 + s_z^2} \right] = \mathcal{L}_{jk}^{\text{spec}}(Z_j), \quad (1)$$

where the second argument is the location and the third is the scale. The function $b_G(x)$ is a magnitude-dependent *relative* bias, s_z is one common additional scale per star (hence a contribution $2s_z^2$ to a binary difference variance), and ν is the shared degrees of freedom. Conditional on Z_j , the two component measurements are independent.

3.2 Stage 1b: Absolute calibration and system metallicity posteriors

3.2.1 CMD likelihood

In principle, we can make use of any set of anchor points whose metallicities are robust and cover a wide enough range to get the absolute calibration.

Here’s an example when we directly use the PARSEC model points of warm main-sequence stars. The PARSEC surface supplies predicted colour $C_P(x, Z)$ and effective temperature $T_{\text{eff},P}(x, Z)$ (Bressan et al., 2012). The repository records the age, metallicity, and magnitude coverage of the local PARSEC table, but not enough provenance to identify a more specific PARSEC release or Gaia passband realization; we therefore make no more specific version claim. The formal colour uncertainty is propagated from the BP and RP signal-to-noise ratios,

$$\sigma_{C,jk} = \frac{2.5}{\ln 10} \left(\text{SNR}_{\text{BP},jk}^{-2} + \text{SNR}_{\text{RP},jk}^{-2} \right)^{1/2}. \quad (2)$$

For an eligible component, the CMD likelihood is

$$C_{jk} | Z_j \sim \mathcal{T}_4 \left[C_P(x_{jk}, Z_j), \sqrt{\sigma_{C,jk}^2 + s_C^2} \right] = \mathcal{L}_{jk}^{\text{CMD}}(Z_j). \quad (3)$$

The colour zero point is fixed to zero and $s_C \sim \text{HalfNormal}(0.1 \text{ mag})$. To avoid allowing membership in the CMD term to change with a proposed metallicity, eligibility is fixed over the full latent-metallicity domain:

$$m_{jk} = \mathbb{I} \left[\min_{q \in \mathcal{Q}} T_{\text{eff},P}(x_{jk}, Z_q) \geq 4000 \text{ K} \right]. \quad (4)$$

If $m_{jk} = 0$, that star still contributes its JCAPS likelihood but not its CMD likelihood. To construct the fixed CMD reference surface, we selected PARSEC rows with $9.3 < \log_{10}(\text{age/yr}) < 10.0$ and $3 < G_{\text{mag}} < 15$. At fixed source-table G_{mag} and metallicity, rows from the selected ages were averaged separately in colour and $\log T_{\text{eff}}$, so age is not a latent variable in this analysis. Colour was represented by an up-to-cubic smoothing spline with smoothing parameter 0.001; $\log T_{\text{eff}}$ was linearly interpolated and then exponentiated before tabulation. The colour and linear- T_{eff} tables were evaluated bilinearly in (M_G, Z) on a 500-point magnitude grid spanning $3.5 \leq M_G \leq 13.5$. This CMD construction is distinct from the 501×81 (M_G, Z) grid used only to fit the MLR’s fixed PARSEC mass projection.

The frozen sample contains 14,876 binaries (29,752 stars). The 4000-K rule admits 12,167 stars; 8,247 systems have at least one eligible component and 3,920 have two. These counts describe CMD coverage, not evidence that the cool-star PARSEC colours are correct.

3.2.2 System posterior and the cut object

The sampled Stage-1a and Stage-1b global parameters are $\psi = (b_G, s_z, \delta_z, s_C)$. The unnormalized metallicity density for one system is

$$\tilde{p}_j(Z, \psi | \mathcal{D}) = p_{\text{pop}}(Z) p(\psi) \prod_{k=1}^2 \mathcal{L}_{jk}^{\text{spec}}(D_{1a} | Z, \psi) \left[\mathcal{L}_{jk}^{\text{CMD}}(D_{1b} | \psi) \right]^{m_{jk}}. \quad (5)$$

The global parameters are inferred from the product of the system-level marginal evidences, rather than from separately normalized system posteriors.

The normalized system posterior is

$$p_j(Z | \mathcal{D}) = \frac{\tilde{p}_j(Z, \psi | \mathcal{D})}{\int \tilde{p}_j(Z, \psi | \mathcal{D}) dZ}. \quad (6)$$

4 Stage 2: a PARSEC-relative hard-monotone MLR

4.1 Baseline projection and scientific correction

Let the original PARSEC mass surface be

$$g_P(x, Z) = \log_{10} \left[\frac{M_P(x, Z)}{M_\odot} \right]. \quad (7)$$

The finite MLR basis cannot reproduce this tabulated surface exactly. The projected PARSEC mass surface onto the same tensor-product B-spline basis used by the fitted model is represented as $g_P^*(x, Z)$,

$$g_P^*(x, Z) = \sum_{i=1}^{K_x} \sum_{h=1}^{K_Z} \Pi_{ih} B_i(x) C_h(Z), \quad (8)$$

where Π is fixed. The fitted correction is

$$\Delta_*(x, Z) = \sum_{i,h} D_{ih} B_i(x) C_h(Z), \quad \Theta_{ih} = \Pi_{ih} + D_{ih}, \quad (9)$$

and hence the fitted mass surface is

$$g(x, Z) = \sum_{i,h} \Theta_{ih} B_i(x) C_h(Z) = g_P^*(x, Z) + \Delta_*(x, Z). \quad (10)$$

The quantity reported scientifically relative to the original PARSEC table is not Δ_* alone but

$$\Delta_P(x, Z) = g(x, Z) - g_P(x, Z) = \Delta_*(x, Z) + \epsilon_P(x, Z), \quad \epsilon_P = g_P^* - g_P. \quad (11)$$

where ϵ_P is the deterministic projection error, the RMS and maximum absolute values of ϵ_P are 0.002 and 0.019 dex, respectively.

The adopted basis has $K_x \times K_Z = 8 \times 4$ cubic B-splines (de Boor, 2001). Both knot vectors are open and clamped:

$$\kappa_x = (3.5, 3.5, 3.5, 3.5, 5.5, 7.5, 9.5, 11.5, 13.5, 13.5, 13.5, 13.5) \text{ mag}, \quad (12)$$

$$\kappa_Z = (-1.0, -1.0, -1.0, -1.0, 0.6, 0.6, 0.6, 0.6) \text{ dex}. \quad (13)$$

4.2 Hard monotonicity of the total mass surface

At fixed metallicity, increasing M_G means a fainter star and must not increase its mass. At fixed magnitude, the adopted physical assumption is that increasing metallicity must not decrease the mass. We therefore require

$$\Theta_{i+1,h} \leq \Theta_{ih}, \quad \Theta_{i,h+1} \geq \Theta_{ih}. \quad (14)$$

Because B-spline derivatives are non-negative lower-degree basis functions weighted by adjacent control-coefficient differences, these inequalities imply throughout the continuous domain

$$\frac{\partial g}{\partial x} \leq 0, \quad \frac{\partial g}{\partial Z} \geq 0. \quad (15)$$

The constraints apply to the *total* coefficient surface $\Theta = \Pi + D$, not to the correction D alone. The correction is consequently free to increase or decrease in either coordinate. We also impose no sign constraint on its mixed differences; doing so would add an unsupported assumption about how the magnitude slope changes with metallicity. The constructive parameterization that enforces equation (14) is given in appendix C.

5 Wide-binary dynamical likelihood

5.1 Mass-scaled velocity statistic

In binary orbital motion, the system mass ($m_{\text{tot}} = m_1 + m_2$), component separation (a), true anomaly θ and eccentricity e set the instantaneous relative speed,

$$v_{\text{rel}} = \sqrt{\frac{G(m_1 + m_2)}{a}} f(\theta, e), \quad (16)$$

with $f(\theta, e)$ encapsulating the phase dependence along the ellipse. At pericenter ($\theta = 0$),

$$v_{\text{rel,peri}} = \sqrt{\frac{G(m_1 + m_2)}{a}} \sqrt{\frac{1+e}{1-e}}, \quad (17)$$

and at apocenter ($\theta = \pi$),

$$v_{\text{rel,apo}} = \sqrt{\frac{G(m_1 + m_2)}{a}} \sqrt{\frac{1-e}{1+e}}. \quad (18)$$

To connect physics with observables we define

$$u := v\sqrt{s}, \quad (19)$$

where v is the projected relative speed and s is the projected separation. Under this definition, $u \propto \sqrt{m_{\text{tot}}/M_{\odot}}$ and depends on nuisance parameters θ , e , and orientation ϕ . We again separate the mass dependence by defining $\tilde{u} \equiv u/\sqrt{m_{\text{tot}}/M_{\odot}}$, so $\tilde{u}$ depends only on the nuisance parameters and their influence can be marginalized. We perform simulations and pre-fit $p(\tilde{u})$ by integrating $p(\tilde{u} | \phi, \theta, e)$ over the nuisance distributions. For an observed u the likelihood then becomes

$$p(u | m_{\text{tot}}) = \frac{1}{\sqrt{m_{\text{tot}}/M_{\odot}}} p\left(\tilde{u} \equiv \frac{u}{\sqrt{m_{\text{tot}}/M_{\odot}}}\right). \quad (20)$$

Our goal is to constrain the mass–luminosity mapping $\mathcal{M}$, which links photometric observables to stellar mass. We treat

$$\{M_{\text{G}}, \text{BP} - \text{RP}, Z\} = \mathcal{M}(m, Z), \quad (21)$$

with parameters Θ defining the functional form. Conversely, stellar masses are obtained from the inverse map $m = \mathcal{M}^{-1}(M_{\text{G}}, Z; \Theta)$. The likelihood for one measurement is then

$$\begin{aligned} \mathcal{L}_j & \\ &= p\left(u_j | m_{\text{tot}}; \mathcal{M}^{-1}(M_{\text{G},j1}, Z_j; \Theta) + \mathcal{M}^{-1}(M_{\text{G},j2}, Z_j; \Theta)\right). \end{aligned} \quad (22)$$

Measurement uncertainties on the tangential velocity enter mainly through the uncertainty of the proper-motion difference $\Delta\mu$ in the relation of

$$u = 4.74 \times \frac{\Delta\mu}{\varpi} \times \sqrt{s}. \quad (23)$$

Assuming Rice-distributed proper-motion differences,

$$\begin{aligned} \text{Rice}(\Delta\mu_{\text{obs}} | \Delta\mu_{\text{true}}, \sigma_{\Delta\mu}) &= \frac{\Delta\mu_{\text{obs}}}{\sigma_{\Delta\mu}^2} \\ &\times \exp\left[-\frac{\Delta\mu_{\text{obs}}^2 + \Delta\mu_{\text{true}}^2}{2\sigma_{\Delta\mu}^2}\right] I_0\left(\frac{\Delta\mu_{\text{obs}} \Delta\mu_{\text{true}}}{\sigma_{\Delta\mu}^2}\right), \end{aligned} \quad (24)$$

the distribution for u is

$$\begin{aligned} \text{Rice}(u_j | u_{\text{true}}, \sigma_{u,j}) &= \frac{u_j}{\sigma_{u,j}^2} \\ &\times \exp\left[-\frac{u_j^2 + u_{\text{true}}^2}{2\sigma_{u,j}^2}\right] I_0\left(\frac{u_j u_{\text{true}}}{\sigma_{u,j}^2}\right), \end{aligned} \quad (25)$$

where I_0 is the zeroth-order modified Bessel function. Integrating over $\tilde{u}$ yields

$$\begin{aligned} p(u_j | m_{\text{tot}}) \\ = \frac{1}{\sqrt{m_{\text{tot}}/M_{\odot}}} \int_0^{\infty} \text{Rice}\left(\frac{u_j}{\sqrt{m_{\text{tot}}/M_{\odot}}} \middle| \tilde{u}, \frac{\sigma_{u,j}}{\sqrt{m_{\text{tot}}/M_{\odot}}}\right) p(\tilde{u}) d\tilde{u}. \end{aligned} \quad (26)$$

To allow for outliers we adopt a mixture model for each datum,

$$\begin{aligned} \mathcal{L}_j^{\text{mix}} &= \int \left[(1 - f_{\text{outlier}}) \cdot \frac{p(\tilde{u})}{\sqrt{m_{\text{tot}}/M_{\odot}}} \right. \\ &\quad \left. + f_{\text{outlier}} \cdot \frac{G_{\text{o}}(\sqrt{m_{\text{tot}}} \cdot \tilde{u} | \mu_{\text{o}}, \sigma_{\text{o}}^2)}{C_{\text{o}}} \right] \\ &\quad \times \text{Rice}\left(\frac{u_{\text{obs}}}{\sqrt{m_{\text{tot}}/M_{\odot}}} \middle| \tilde{u}, \frac{\sigma_{u,j}}{\sqrt{m_{\text{tot}}/M_{\odot}}}\right) d\tilde{u}, \end{aligned} \quad (27)$$

where the outlier fraction f_{outlier} is a free parameter, and G_{o} is a Gaussian distribution with mean μ_{o} and variance σ_{o}^2 (defined on velocity scale u). The normalization constant C_{o} ensures that the outlier distribution integrates to unity over the relevant range of $\tilde{u}$.

The mass-luminosity relation can thus be constrained by substituting m_{tot} with

$$m_{\text{tot},j} = \mathcal{M}^{-1}(M_{\text{G},j1}, \mathbb{E}[Z_j]; \Theta) + \mathcal{M}^{-1}(M_{\text{G},j2}, \mathbb{E}[Z_j]; \Theta), \quad (28)$$

where $\mathbb{E}[Z_j]$ is the expected value of the metallicity posterior inferred from Stage 1b.

$$\mathbb{E}[Z_j] = \int Z \cdot p_j(Z | \mathcal{D}) dZ \quad (29)$$

Appendix

A Parameters, priors, and posterior computation

The sampled MLR coordinates are

$$\boldsymbol{\theta}_M = \left(c_0, \{a_h\}_{h=2}^{K_Z}, \{b_i\}_{i=2}^{K_x}, \{r_{ih}\}_{i,h \geq 2}, \log \lambda_x, \log \lambda_Z, f_{\text{outlier}} \right), \quad (30)$$

giving 35 scalar degrees of freedom for $K_x = 8$ and $K_Z = 4$. The priors and fixed regularization scales are summarized in [table 2](#). Here $\Pi_{11} = 0.00703$, and the PARSEC-derived reference increments are $\lambda_{x,\text{ref}} = 0.18624$ and $\lambda_{Z,\text{ref}} = 0.13078$ dex.

Table 2: Frozen Stage-1a, Stage-1b, and Stage-2 parameter specification. Normal distributions are written as location and standard deviation.

Stage	Quantity	Prior or fixed value	Role
1a	$\boldsymbol{\theta}_{\text{rel}}$	MLE; no prior	Difference calibration (b_G, s_z, ν); fixed in Stage 1b
1b	δ_z	$\mathcal{N}(0, 0.3)$ dex	JCAPS–PARSEC zero point
1b	s_C	Half-Normal(0.1) mag	CMD excess colour scatter
1b	$\boldsymbol{\omega}$	Dirichlet(1, ..., 1)	Six-component population weights
2	c_0	$\mathcal{N}(\Pi_{11}, 0.10)$ dex	First total-surface coefficient
2	a_h, b_i, r_{ih}	$\mathcal{N}(0, 1)$	Unconstrained monotone coordinates
2	$\log \lambda_x$	$\mathcal{N}(\log 0.18624, 0.7)$	Magnitude increment scale
2	$\log \lambda_Z$	$\mathcal{N}(\log 0.13078, 0.7)$	Metallicity increment scale
2	f_{outlier}	Beta(3, 12)	Effective component weight

Stage 1b and Stage 2 were sampled with the No-U-Turn Sampler (NUTS) ([Hoffman and Gelman, 2014](#)), using four chains and 1,000 retained draws per chain. The frozen fits had no divergent transitions. Posterior adequacy must additionally be assessed with rank-based $\hat{R}$, effective sample sizes, chain traces, and posterior predictive or held-out checks; successful execution alone is not scientific validation.

B Information propagated, assumptions, and limitations

[Table 3](#) distinguishes uncertainty that is represented in the current likelihood from uncertainty held fixed or omitted. In particular, the model propagates the reported per-star JCAPS errors, heavy-tailed metallicity scatter, CMD colour scatter, the single-system metallicity posterior, and the reported dynamical amplitude error. It does not propagate uncertainty in the absolute magnitudes, extinction correction, PARSEC model family, age distribution, or uncertainty in the preliminary Stage-1a relative-calibration estimates.

Table 3: Uncertainty and information flow in the frozen model.

Category	Represented	Not represented or deliberately cut
Metallicity	Raw per-star errors, shared Z_j , Student- t tails, CMD excess scatter, population weights, and each system's $\bar{P}_{jq}$	Uncertainty in the fixed preliminary Stage-1a estimates b_G , s_z , and ν ; cross-system covariance induced by shared Stage-1b parameters
Photometry and models	Formal BP/RP colour errors and fitted s_C	M_G uncertainty, extinction uncertainty, age as a latent variable, PARSEC-family/model discrepancy
Dynamics	Rice convolution with reported $\sigma_{u,j}$ and latent velocity component distributions	Uncertainty in the fixed good/outlier shapes and additional astrometric covariance beyond the reported $\sigma_{u,j}$
Feedback	Metallicity uncertainty is passed from Stage 1b to Stage 2	Dynamics cannot update δ_z , s_C , ω , or individual metallicity posteriors

Several interpretation boundaries follow. First, the shared Z_j is a model assumption for a co-natal binary, not direct proof that both stars have identical surface abundances. Second, the 4000-K threshold expands CMD coverage but may also increase sensitivity to low-temperature PARSEC colour systematics. Third, hard monotonicity encodes the adopted physical ordering of the MLR; it does not demonstrate that PARSEC or the observations are unbiased over the full domain. Fourth, the cut reduces feedback from a potentially misspecified dynamical model, but it understates joint uncertainty by fixing the Stage-1a estimates, discarding the shared Stage-1b posterior covariance, and excluding any information that the dynamics could provide about metallicity calibration. A fully joint analysis would be a different model, not merely a more accurate implementation of the present one.

C Constructive monotone reparameterization

The total coefficient grid Θ is generated from unconstrained raw variables rather than by rejecting proposals that violate inequalities. Define $\text{softplus}(t) = \log(1 + e^t)$ and $\text{sigmoid}(t) = (1 + e^{-t})^{-1}$. With positive scales $\lambda_x = e^{\log \lambda_x}$ and $\lambda_Z = e^{\log \lambda_Z}$, construct the first row and column as

$$\Theta_{11} = c_0, \quad (31)$$

$$\Theta_{1h} = \Theta_{1,h-1} + \lambda_Z \text{softplus}(a_h), \quad h = 2, \dots, K_Z, \quad (32)$$

$$\Theta_{i1} = \Theta_{i-1,1} - \lambda_x \text{softplus}(b_i), \quad i = 2, \dots, K_x. \quad (33)$$

The first row therefore increases with Z , whereas the first column decreases with x . Interior points are visited in increasing $i + h$. For each point, define

$$L_{ih} = \Theta_{i,h-1}, \quad U_{ih} = \Theta_{i-1,h}, \quad (34)$$

and set

$$\Theta_{ih} = L_{ih} + (U_{ih} - L_{ih}) \text{sigmoid}(r_{ih}), \quad i, h \geq 2. \quad (35)$$

If all previously constructed points obey the two monotone orders, then

$$L_{ih} = \Theta_{i,h-1} \leq \Theta_{i-1,h-1} \leq \Theta_{i-1,h} = U_{ih}. \quad (36)$$

Thus [equation \(35\)](#) lies inside a non-empty interval and simultaneously satisfies $\Theta_{i,h-1} \leq \Theta_{ih} \leq \Theta_{i-1,h}$. Induction over $i + h$ therefore proves the coefficient inequalities in [equation \(14\)](#) for the complete grid. Finally, $D = \Theta - \Pi$ is computed deterministically.

For completeness, the derivative identity for a degree- d_x B-spline surface has the form

$$\frac{\partial g}{\partial x} = \sum_{i=1}^{K_x-1} \sum_{h=1}^{K_Z} \rho_{x,i} (\Theta_{i+1,h} - \Theta_{ih}) \tilde{B}_i(x) C_h(Z), \quad \rho_{x,i} > 0, \quad (37)$$

where $\tilde{B}_i$ is a non-negative degree- $(d_x - 1)$ basis function. The corresponding Z derivative contains $\rho_{Z,h} (\Theta_{i,h+1} - \Theta_{ih}) B_i \tilde{C}_h$ with $\rho_{Z,h} > 0$. The coefficient orders therefore establish the derivative signs everywhere in the spline domain, including the appropriate one-sided limits at its boundaries.

References

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